



Exploratory Data Analysis (EDA) Questions

Complete the following tasks using pandas, NumPy, and visualization libraries. Write your own code in the cells provided below. Do not include answers; only solve each question by writing code.

Setup

Import the required Python libraries for data analysis and visualization.

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

Load the Dataset

Load the provided CSV dataset into a pandas DataFrame and display the first five records to verify that it has been loaded correctly.

```
In [4]: df = pd.read_csv(r"C:\Users\uktiw\Downloads\ai_workforce_displacement_global_2020.csv")
df.head()
```

```
Out[4]:
```

	record_id	country	iso3_code	region	income_group	year	quarter	quarter
0	1	United States	USA	North America	High Income	2020	1	2020
1	2	United States	USA	North America	High Income	2020	1	2020
2	3	United States	USA	North America	High Income	2020	1	2020
3	4	United States	USA	North America	High Income	2020	1	2020
4	5	United States	USA	North America	High Income	2020	1	2020

5 rows × 23 columns

Question 1: Dataset Overview

Perform an initial exploration of the dataset. Write code to determine:

- Number of rows and columns
- Data types of all columns
- Summary statistics for numerical features
- Number of unique values in each column
- Missing values in each column
- Duplicate records in the dataset.

```
In [10]: df.value_counts
```

```
Out[10]: <bound method DataFrame.value_counts of
3_code      region      income_group \
0           1  United States      USA  North America      High Income
1           2  United States      USA  North America      High Income
2           3  United States      USA  North America      High Income
3           4  United States      USA  North America      High Income
4           5  United States      USA  North America      High Income
...
20795      20796      Zambia      ZMB      Africa  Lower Middle Income
20796      20797      Zambia      ZMB      Africa  Lower Middle Income
20797      20798      Zambia      ZMB      Africa  Lower Middle Income
20798      20799      Zambia      ZMB      Africa  Lower Middle Income
20799      20800      Zambia      ZMB      Africa  Lower Middle Income
```

```

year  quarter  quarter_label      industry_sector \
0    2020      1      2020-Q1      Technology & Software
1    2020      1      2020-Q1      Finance & Banking
2    2020      1      2020-Q1      Healthcare & Life Sciences
3    2020      1      2020-Q1      Manufacturing & Industry
4    2020      1      2020-Q1      Retail & E-Commerce
...
20795  2026      2      2026-Q2      Education & Research
20796  2026      2      2026-Q2      Transportation & Logistics
20797  2026      2      2026-Q2      Media & Communications
20798  2026      2      2026-Q2      Administrative & Clerical
20799  2026      2      2026-Q2      Energy & Utilities
```

```

sector_automation_risk_score ... \
0           0.382 ...
1           0.608 ...
2           0.198 ...
3           0.720 ...
4           0.676 ...
...
20795      0.308 ...
20796      0.675 ...
20797      0.516 ...
20798      0.801 ...
20799      0.440 ...
```

```

pct_sector_workforce_new_roles_created  net_workforce_change_pct \
0           0.0348           -0.0058
1           0.0423           -0.0094
2           0.0168           -0.0008
3           0.0642           -0.0282
4           0.0374           -0.0293
...
20795      0.0056           -0.0162
20796      0.0205           -0.0329
20797      0.0154           -0.0351
20798      0.0260           -0.0654
20799      0.0160           -0.0216
```

```
ai_cited_layoff_announcements  ai_skill_wage_premium_pct \
```

0	32	0.311
1	51	0.304
2	4	0.278
3	47	0.268
4	31	0.313
...	...	...
20795	29	0.450
20796	126	0.476
20797	131	0.446
20798	476	0.499
20799	67	0.490

	pct_workforce_female	pct_displaced_roles_female \
0	0.28	0.275
1	0.48	0.511
2	0.72	0.707
3	0.29	0.310
4	0.54	0.547
...	...	...
20795	0.58	0.651
20796	0.22	0.235
20797	0.46	0.482
20798	0.79	0.950
20799	0.26	0.270

	reskilling_programs_count	govt_ai_policy_score_1_to_10 \
0	52	6.4
1	39	6.5
2	37	6.8
3	50	6.8
4	37	6.2
...	...	...
20795	16	3.5
20796	15	4.2
20797	17	4.3
20798	14	4.2
20799	12	4.1

	ai_tool_adoption_pct	data_source_note
s		
0	0.280	Research-calibrated synthetic data. Grounded
i...		
1	0.559	Research-calibrated synthetic data. Grounded
i...		
2	0.202	Research-calibrated synthetic data. Grounded
i...		
3	0.655	Research-calibrated synthetic data. Grounded
i...		
4	0.542	Research-calibrated synthetic data. Grounded
i...		
...	...	
...		
20795	0.184	Research-calibrated synthetic data. Grounded
i...		

```

20796          0.457  Research-calibrated synthetic data. Grounded
i...
20797          0.419  Research-calibrated synthetic data. Grounded
i...
20798          0.459  Research-calibrated synthetic data. Grounded
i...
20799          0.377  Research-calibrated synthetic data. Grounded
i...

```

```
[20800 rows x 23 columns]>
```

```
In [11]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20800 entries, 0 to 20799
Data columns (total 23 columns):
#   Column                                          Non-Null Count  Dtype
---  -
0   record_id                                     20800 non-null  int64
1   country                                       20800 non-null  object
2   iso3_code                                     20800 non-null  object
3   region                                       20800 non-null  object
4   income_group                                 20800 non-null  object
5   year                                         20800 non-null  int64
6   quarter                                       20800 non-null  int64
7   quarter_label                                20800 non-null  object
8   industry_sector                             20800 non-null  object
9   sector_automation_risk_score                20800 non-null  float64
10  gdp_per_capita_usd                          20800 non-null  int64
11  ai_adoption_index                           20800 non-null  float64
12  pct_sector_workforce_displaced              20800 non-null  float64
13  pct_sector_workforce_new_roles_created      20800 non-null  float64
14  net_workforce_change_pct                    20800 non-null  float64
15  ai_cited_layoff_announcements               20800 non-null  int64
16  ai_skill_wage_premium_pct                   20800 non-null  float64
17  pct_workforce_female                        20800 non-null  float64
18  pct_displaced_roles_female                  20800 non-null  float64
19  reskilling_programs_count                   20800 non-null  int64
20  govt_ai_policy_score_1_to_10                20800 non-null  float64
21  ai_tool_adoption_pct                        20800 non-null  float64
22  data_source_notes                           20800 non-null  object
dtypes: float64(10), int64(6), object(7)
memory usage: 3.7+ MB

```

```
In [12]: df.describe()
```

Out[12]:	record_id	year	quarter	sector_automation_risk_score
count	20800.000000	20800.000000	20800.000000	20800.000000
mean	10400.500000	2022.769231	2.423077	0.537881
std	6004.587135	1.887406	1.115411	0.190345
min	1.000000	2020.000000	1.000000	0.151000
25%	5200.750000	2021.000000	1.000000	0.379000
50%	10400.500000	2023.000000	2.000000	0.580000
75%	15600.250000	2024.000000	3.000000	0.698000
max	20800.000000	2026.000000	4.000000	0.855000

In [13]: `df.isnull().sum()`

```
Out[13]: record_id          0
country          0
iso3_code        0
region           0
income_group     0
year             0
quarter          0
quarter_label    0
industry_sector  0
sector_automation_risk_score  0
gdp_per_capita_usd  0
ai_adoption_index  0
pct_sector_workforce_displaced  0
pct_sector_workforce_new_roles_created  0
net_workforce_change_pct  0
ai_cited_layoff_announcements  0
ai_skill_wage_premium_pct  0
pct_workforce_female  0
pct_displaced_roles_female  0
reskilling_programs_count  0
govt_ai_policy_score_1_to_10  0
ai_tool_adoption_pct  0
data_source_notes  0
dtype: int64
```

In [14]: `df.nunique()`

```
Out[14]: record_id          20800
country          80
iso3_code        80
region           12
income_group      4
year             7
quarter          4
quarter_label     26
industry_sector   10
sector_automation_risk_score  679
gdp_per_capita_usd 1981
ai_adoption_index 775
pct_sector_workforce_displaced 2068
pct_sector_workforce_new_roles_created 1680
net_workforce_change_pct 837
ai_cited_layoff_announcements 519
ai_skill_wage_premium_pct 490
pct_workforce_female 10
pct_displaced_roles_female 453
reskilling_programs_count 91
govt_ai_policy_score_1_to_10 78
ai_tool_adoption_pct 799
data_source_notes 1
dtype: int64
```

```
In [17]: df.duplicated().sum()
```

```
Out[17]: np.int64(0)
```

Question 2

Count the number of unique countries.

```
In [21]: df['country'].nunique()
```

```
Out[21]: 80
```

Question 3

Display all unique industry sectors.

```
In [22]: df['industry_sector'].unique()
```

```
Out[22]: array(['Technology & Software', 'Finance & Banking',
                'Healthcare & Life Sciences', 'Manufacturing & Industry',
                'Retail & E-Commerce', 'Education & Research',
                'Transportation & Logistics', 'Media & Communications',
                'Administrative & Clerical', 'Energy & Utilities'], dtype=object)
```

Question 4

Find the average AI Adoption Index.

```
In [23]: df['ai_adoption_index'].mean()
```

```
Out[23]: np.float64(0.6921477884615385)
```

Question 5

Which country has the highest GDP per capita?

```
In [29]: df.groupby('country')['gdp_per_capita_usd'].mean().sort_values(ascending=False)
```

```
Out[29]: country
Switzerland    90747.153846
Name: gdp_per_capita_usd, dtype: float64
```

Question 6

Find the top 10 countries with the highest AI Adoption Index.

```
In [32]: df.groupby('country')['ai_adoption_index'].mean().sort_values(ascending=False)
```

```
Out[32]:
```

	country	ai_adoption_index
--	---------	-------------------

0	Singapore	0.905346
1	Israel	0.885842
2	United States	0.885408
3	Sweden	0.880223
4	Denmark	0.872588
5	Switzerland	0.872473
6	South Korea	0.868415
7	Finland	0.867404
8	China	0.860900
9	Norway	0.859692

Question 7

Find the average AI Adoption Index by region.

```
In [28]: df.groupby('region')['ai_adoption_index'].mean().reset_index()
```


Out[28]:

	region	ai_adoption_index
0	Africa	0.537141
1	Central Asia	0.609708
2	East Asia	0.848202
3	Europe	0.796663
4	Europe/Asia	0.698483
5	Latin America	0.641559
6	Middle East	0.735563
7	Middle East/Africa	0.574108
8	North America	0.863188
9	Oceania	0.827458
10	South Asia	0.570106
11	Southeast Asia	0.662900

Question 8

Find the average workforce displacement by industry.

In [35]: `df.groupby('industry_sector')['pct_sector_workforce_displaced'].mean().reset_`

Out[35]:

	industry_sector	pct_sector_workforce_displaced
0	Administrative & Clerical	0.092016
1	Education & Research	0.025171
2	Energy & Utilities	0.041974
3	Finance & Banking	0.060348
4	Healthcare & Life Sciences	0.020882
5	Manufacturing & Industry	0.079422
6	Media & Communications	0.054123
7	Retail & E-Commerce	0.073076
8	Technology & Software	0.045995
9	Transportation & Logistics	0.068542

Question 9

Which industry has the highest automation risk?

```
In [40]: df.groupby('industry_sector')['sector_automation_risk_score'].mean().sort_val
```

```
Out[40]: industry_sector  
Administrative & Clerical    0.789742  
Name: sector_automation_risk_score, dtype: float64
```

Question 10

Find countries with AI Adoption Index greater than 0.8.

```
In [50]: df1 = df.groupby('country')['ai_adoption_index'].mean().reset_index()
```

```
In [51]: df1[df1['ai_adoption_index']>0.8]
```

Out[51]:

	country	ai_adoption_index
2	Australia	0.827315
3	Austria	0.817785
5	Belgium	0.822019
10	Canada	0.840969
12	China	0.860900
16	Denmark	0.872588
20	Finland	0.867404
21	France	0.817712
22	Germany	0.834996
26	Hong Kong	0.855296
32	Israel	0.885842
35	Japan	0.809450
44	Netherlands	0.854769
45	New Zealand	0.827600
47	Norway	0.859692
60	Singapore	0.905346
62	South Korea	0.868415
65	Sweden	0.880223
66	Switzerland	0.872473
67	Taiwan	0.846946
72	UAE	0.834019
74	United Kingdom	0.848312
75	United States	0.885408

Question 11

Display only records from 2025./

In [53]: `df[df['year']== 2025]`

Out[53]:	record_id	country	iso3_code	region	income_group	year	quarter	qu
200	201	United States	USA	North America	High Income	2025	1	
201	202	United States	USA	North America	High Income	2025	1	
202	203	United States	USA	North America	High Income	2025	1	
203	204	United States	USA	North America	High Income	2025	1	
204	205	United States	USA	North America	High Income	2025	1	
...	...	...	...	...	...	...	...	...
20775	20776	Zambia	ZMB	Africa	Lower Middle Income	2025	4	
20776	20777	Zambia	ZMB	Africa	Lower Middle Income	2025	4	
20777	20778	Zambia	ZMB	Africa	Lower Middle Income	2025	4	
20778	20779	Zambia	ZMB	Africa	Lower Middle Income	2025	4	
20779	20780	Zambia	ZMB	Africa	Lower Middle Income	2025	4	

3200 rows × 23 columns

Question 12

Find the average wage premium for each income group.

```
In [54]: df.groupby('income_group')['ai_skill_wage_premium_pct'].mean().reset_index()
```

```
Out[54]:
```

	income_group	ai_skill_wage_premium_pct
0	High Income	0.394390
1	Low Income	0.310399
2	Lower Middle Income	0.330390
3	Upper Middle Income	0.356316

Question 13

Count observations for each region.

```
In [55]: df['region'].value_counts()
```

```
Out[55]: region
Europe                5200
Latin America         3380
Africa                3380
Southeast Asia        2080
Middle East           1820
East Asia             1300
South Asia            1300
North America          520
Oceania               520
Europe/Asia           520
Central Asia          520
Middle East/Africa    260
Name: count, dtype: int64
```

Question 14

Find the top five countries with the highest workforce displacement.

```
In [56]: df.groupby('country')['pct_sector_workforce_displaced'].mean().sort_values(asc
```

```
Out[56]: country
Singapore      0.106867
Denmark         0.104123
United States   0.103910
Finland         0.103638
Sweden          0.103595
Name: pct_sector_workforce_displaced, dtype: float64
```

Question 15

Calculate the correlation matrix.

```
In [64]: numeric_df = df.select_dtypes(include='number')
```

```
numeric_df.corr()
```

Out[64]:

	record_id	year	quarter
record_id	1.000000	1.235380e-02	3.732868e-04
year	0.012354	1.000000e+00	-1.180546e-01
quarter	0.000373	-1.180546e-01	1.000000e+00
sector_automation_risk_score	-0.000042	-7.469622e-04	-2.721199e-04
gdp_per_capita_usd	-0.444218	4.806326e-02	3.429363e-04
ai_adoption_index	-0.404061	5.542376e-01	2.853618e-02
pct_sector_workforce_displaced	-0.291737	4.502595e-01	8.420588e-03
pct_sector_workforce_new_roles_created	-0.325253	3.360655e-01	5.721231e-03
net_workforce_change_pct	0.076561	-5.142227e-01	-1.096248e-02
ai_cited_layoff_announcements	-0.077123	5.143983e-01	1.245326e-02
ai_skill_wage_premium_pct	-0.165218	8.778019e-01	-5.913468e-02
pct_workforce_female	0.000023	9.760530e-17	-9.829869e-18
pct_displaced_roles_female	0.000331	-6.459176e-04	6.253342e-04
reskilling_programs_count	-0.379482	2.635605e-01	-3.143848e-02
govt_ai_policy_score_1_to_10	-0.376267	2.175426e-01	-2.539006e-02
ai_tool_adoption_pct	-0.206153	2.882533e-01	1.498766e-02

Question 16

Find the average AI adoption for each year.

```
In [65]: df.groupby('year')['ai_adoption_index'].mean().reset_index()
```

Out[65]:

	year	ai_adoption_index
0	2020	0.542641
1	2021	0.610068
2	2022	0.667465
3	2023	0.717097
4	2024	0.758661
5	2025	0.794465
6	2026	0.817129

Question 17

Find the average net workforce change by industry.

```
In [66]: df.groupby('industry_sector')['net_workforce_change_pct'].mean().reset_index()
```

```
Out[66]:
```

	industry_sector	net_workforce_change_pct
0	Administrative & Clerical	-0.033699
1	Education & Research	-0.009323
2	Energy & Utilities	-0.015178
3	Finance & Banking	-0.022037
4	Healthcare & Life Sciences	-0.007478
5	Manufacturing & Industry	-0.029116
6	Media & Communications	-0.019889
7	Retail & E-Commerce	-0.026680
8	Technology & Software	-0.016672
9	Transportation & Logistics	-0.025166

Question 18

Which region has the highest average government AI policy score?

```
In [67]: df.groupby('region')['govt_ai_policy_score_1_to_10'].mean().sort_values(ascending=False)
```

```
Out[67]: region
North America    7.014615
Name: govt_ai_policy_score_1_to_10, dtype: float64
```

Question 19

Create a new column called AI Readiness Score.

AI Readiness = AI Adoption + Policy Score

```
In [68]: df['ai_readiness_score'] = df['ai_adoption_index'] + df['govt_ai_policy_score_1_to_10']
```

```
Out[68]: 0      7.230
         1      7.317
         2      7.616
         3      7.618
         4      7.028
         ...
        20795    4.203
        20796    4.911
        20797    5.009
        20798    4.902
        20799    4.818
        Name: ai_readiness_score, Length: 20800, dtype: float64
```

Question 20

Display countries where workforce displacement exceeds new roles created.

```
In [88]: df2 = df.groupby('country')['pct_sector_workforce_displaced'].mean().reset_index()
```

```
In [89]: df3 = df.groupby('country')['pct_sector_workforce_new_roles_created'].mean().reset_index()
```

```
In [94]: df4 = pd.merge(df2, df3, how='left')
         df4[df4['pct_sector_workforce_displaced'] > df4['pct_sector_workforce_new_roles_created']]
```

```
Out[94]:
```

	country	pct_sector_workforce_displaced	pct_sector_workforce_new_roles_created
0	Algeria	0.030793	0
1	Argentina	0.041730	0
2	Australia	0.098747	0
3	Austria	0.097350	0
4	Bangladesh	0.028423	0
...	...	...	...
75	United States	0.103910	0
76	Uruguay	0.051993	0
77	Uzbekistan	0.029628	0
78	Vietnam	0.034400	0
79	Zambia	0.025905	0

80 rows × 3 columns

Question 21

Find the average reskilling programs count by income group.

```
In [72]: df.groupby('income_group')['reskilling_programs_count'].mean().reset_index()
```

```
Out[72]:
```

	income_group	reskilling_programs_count
0	High Income	59.348511
1	Low Income	3.457692
2	Lower Middle Income	11.481090
3	Upper Middle Income	28.823427

Question 22

Find the year with the highest average AI Adoption.

```
In [75]: df.groupby('year')['ai_adoption_index'].mean().sort_values(ascending=False).re
```

```
Out[75]:
```

	year	ai_adoption_index
0	2026	0.817129

Question 23

Display the top five records with the highest AI wage premium.

```
In [80]: df[['country', 'industry_sector', 'ai_skill_wage_premium_pct']].sort_values(by=
```

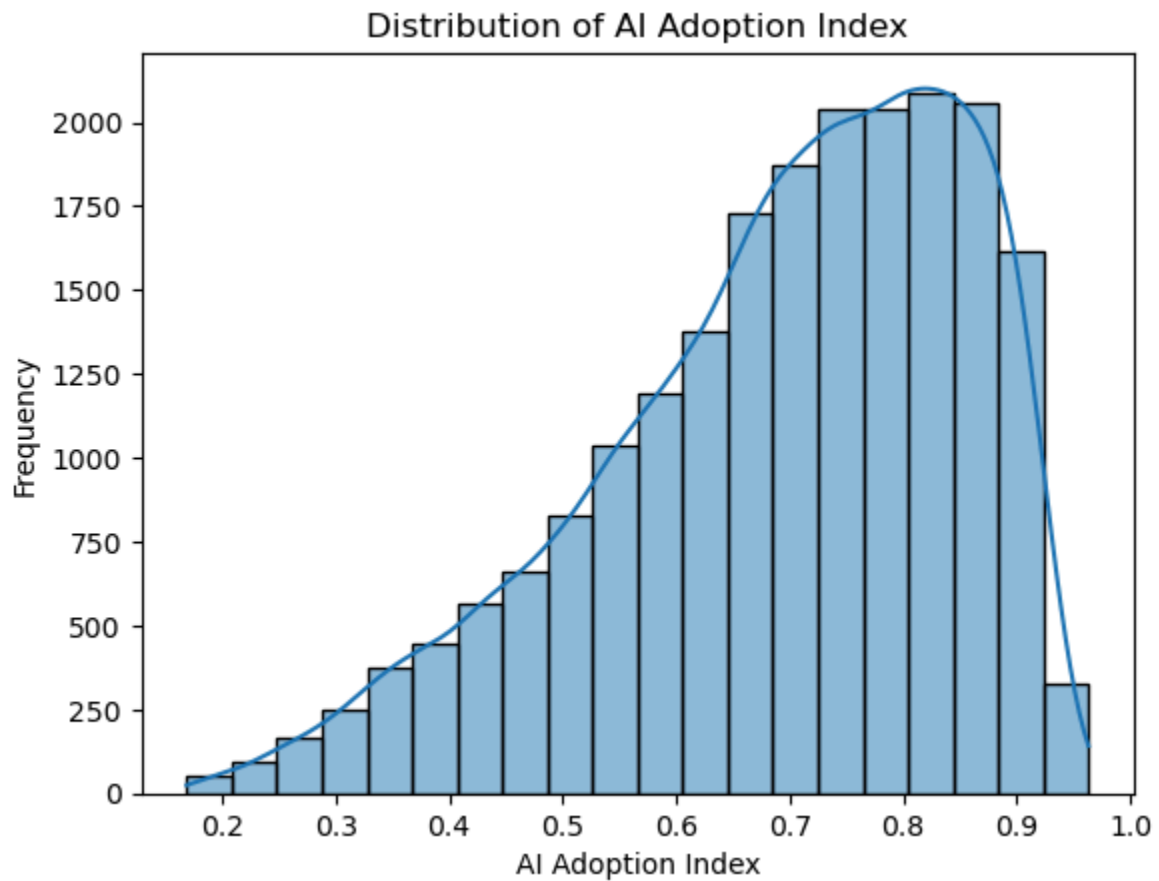
```
Out[80]:
```

	country	industry_sector	ai_skill_wage_premium_pct
5700	Denmark	Technology & Software	0.610
14039	UAE	Energy & Utilities	0.603
16362	Hong Kong	Healthcare & Life Sciences	0.593
5195	Sweden	Education & Research	0.592
12218	Portugal	Administrative & Clerical	0.591

Question 24

Distribution of AI Adoption Index

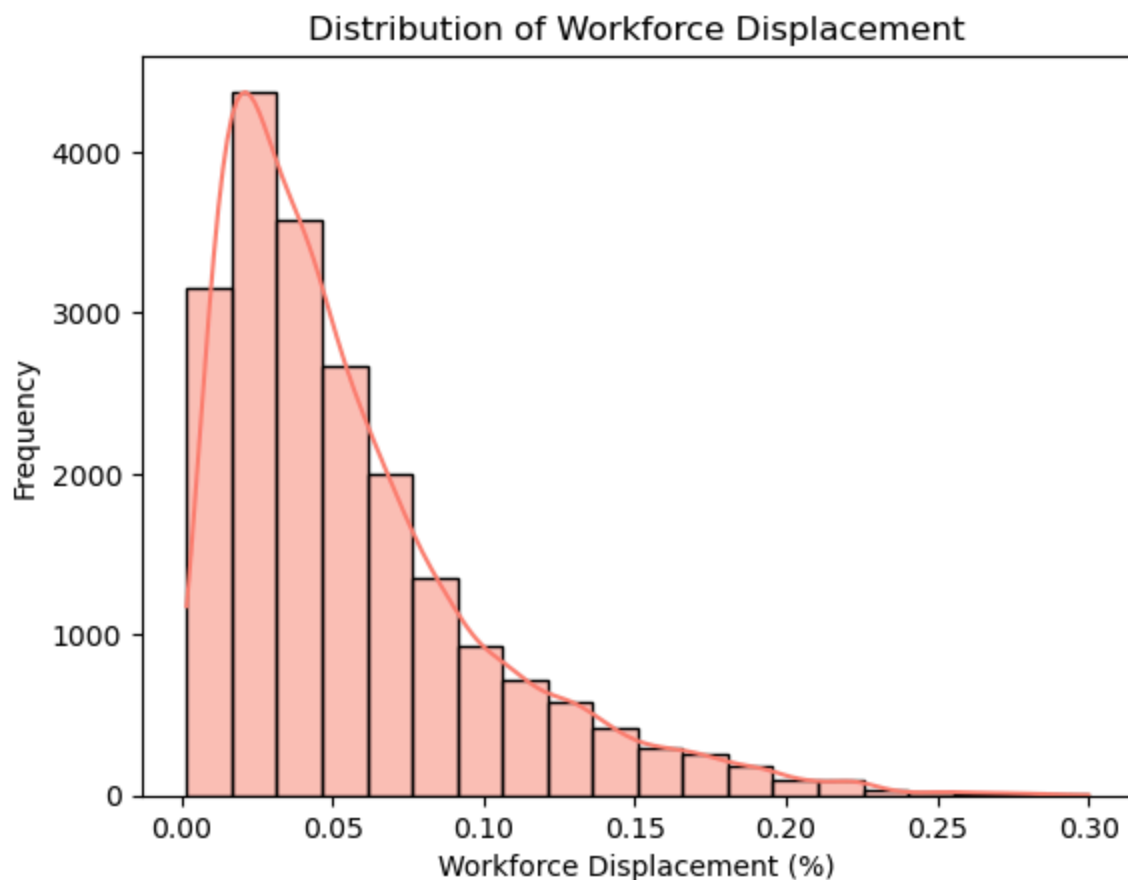
```
In [99]: sns.histplot(df['ai_adoption_index'], bins=20, kde=True)
plt.title("Distribution of AI Adoption Index")
plt.xlabel("AI Adoption Index")
plt.ylabel("Frequency")
plt.show()
```



Question 25

Workforce Displacement Distribution

```
In [100... sns.histplot(df['pct_sector_workforce_displaced'], bins=20, kde=True, color='s')
plt.title("Distribution of Workforce Displacement")
plt.xlabel("Workforce Displacement (%)")
plt.ylabel("Frequency")
plt.show()
```



Question 26

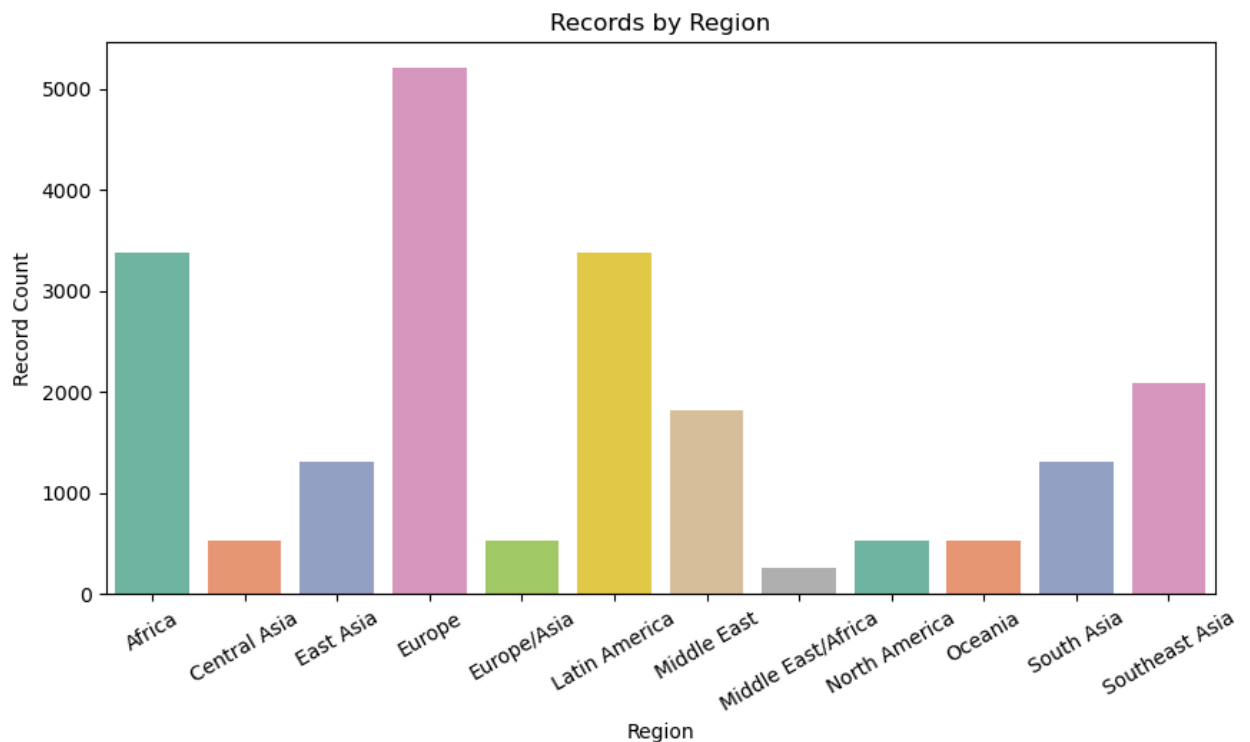
Records by Region

```
In [109... Records_by_Region = (df.groupby('region').size().reset_index(name='record_count'))
plt.figure(figsize = (10,5))
sns.barplot(data=Records_by_Region, x='region', y='record_count', palette="Set2")
plt.title("Records by Region")
plt.xlabel("Region")
plt.ylabel("Record Count")
plt.xticks(rotation=30)
plt.show()
```

C:\Users\uktiw\AppData\Local\Temp\ipykernel_8088\1573519348.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=Records_by_Region, x='region', y='record_count', palette="Set2")
```



Question 27

AI Adoption by Region

```
In [113... ai_by_region = (df.groupby('region')['ai_adoption_index'].mean().reset_index()

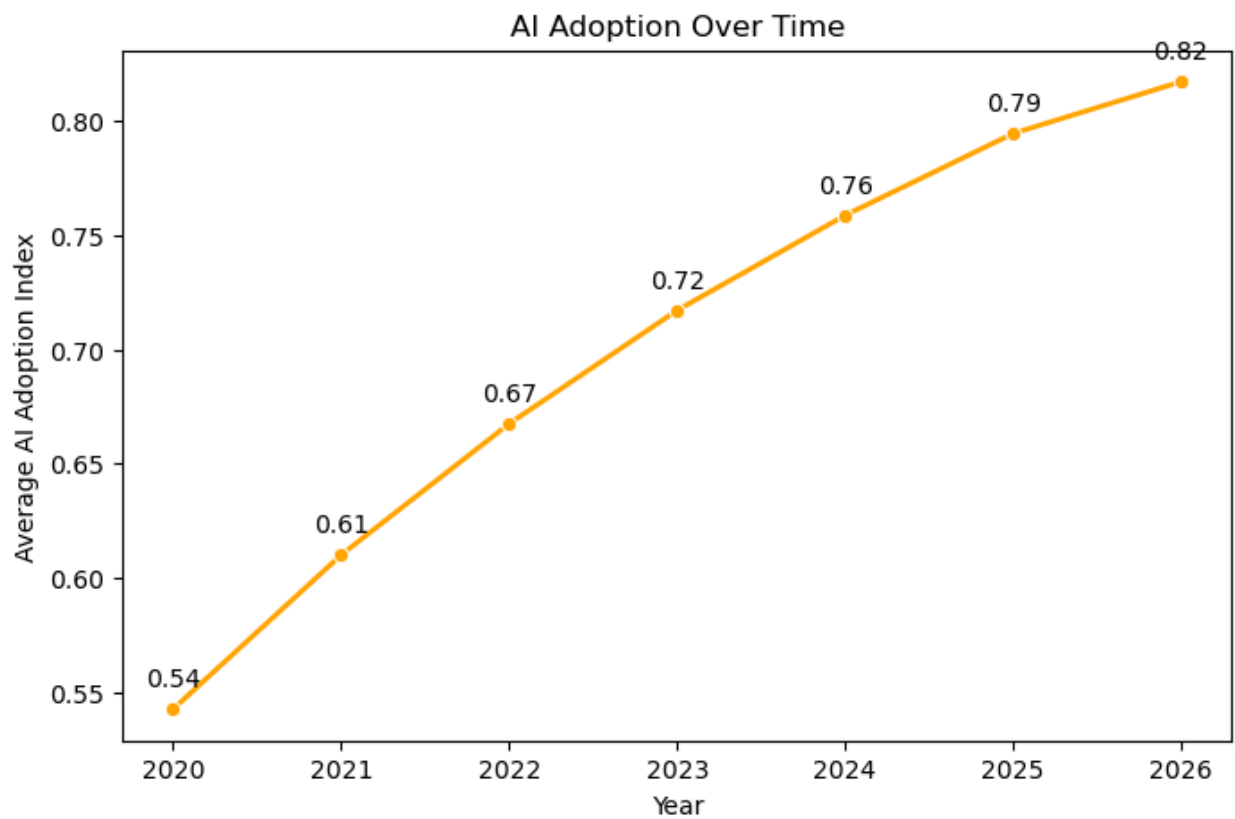
plt.figure(figsize=(8,5))
sns.barplot(data=ai_by_region, x='ai_adoption_index', y='region', palette="vir

plt.title("AI Adoption by Region")
plt.xlabel("Average AI Adoption Index")
plt.ylabel("Region")
plt.show()
```

C:\Users\uktiw\AppData\Local\Temp\ipykernel_8088\2798803372.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

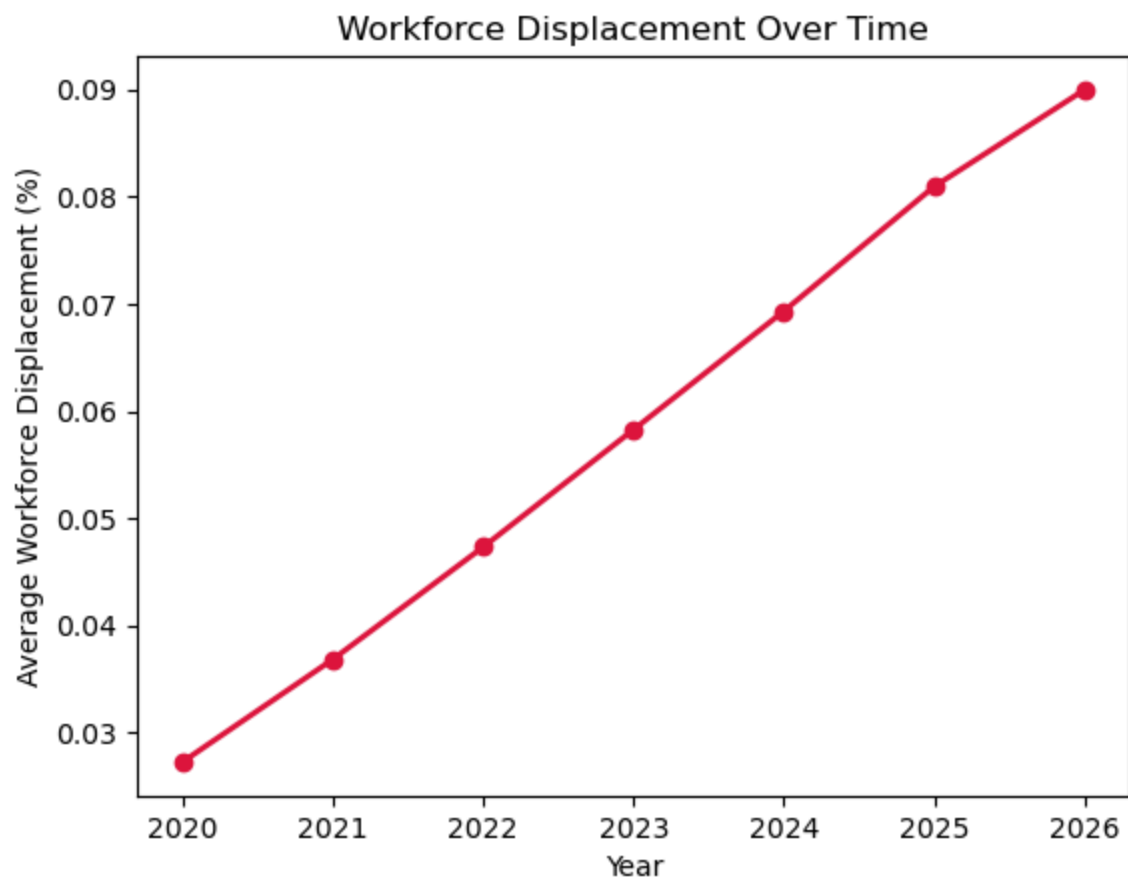
```
sns.barplot(data=ai_by_region, x='ai_adoption_index', y='region', palette="viridis")
```

Question 29

Workforce Displacement Over Time

```
In [121... df.groupby('year')['pct_sector_workforce_displaced'].mean().plot(kind = 'line')  
  
plt.title("Workforce Displacement Over Time")  
plt.xlabel("Year")  
plt.ylabel("Average Workforce Displacement (%)")  
plt.show()
```

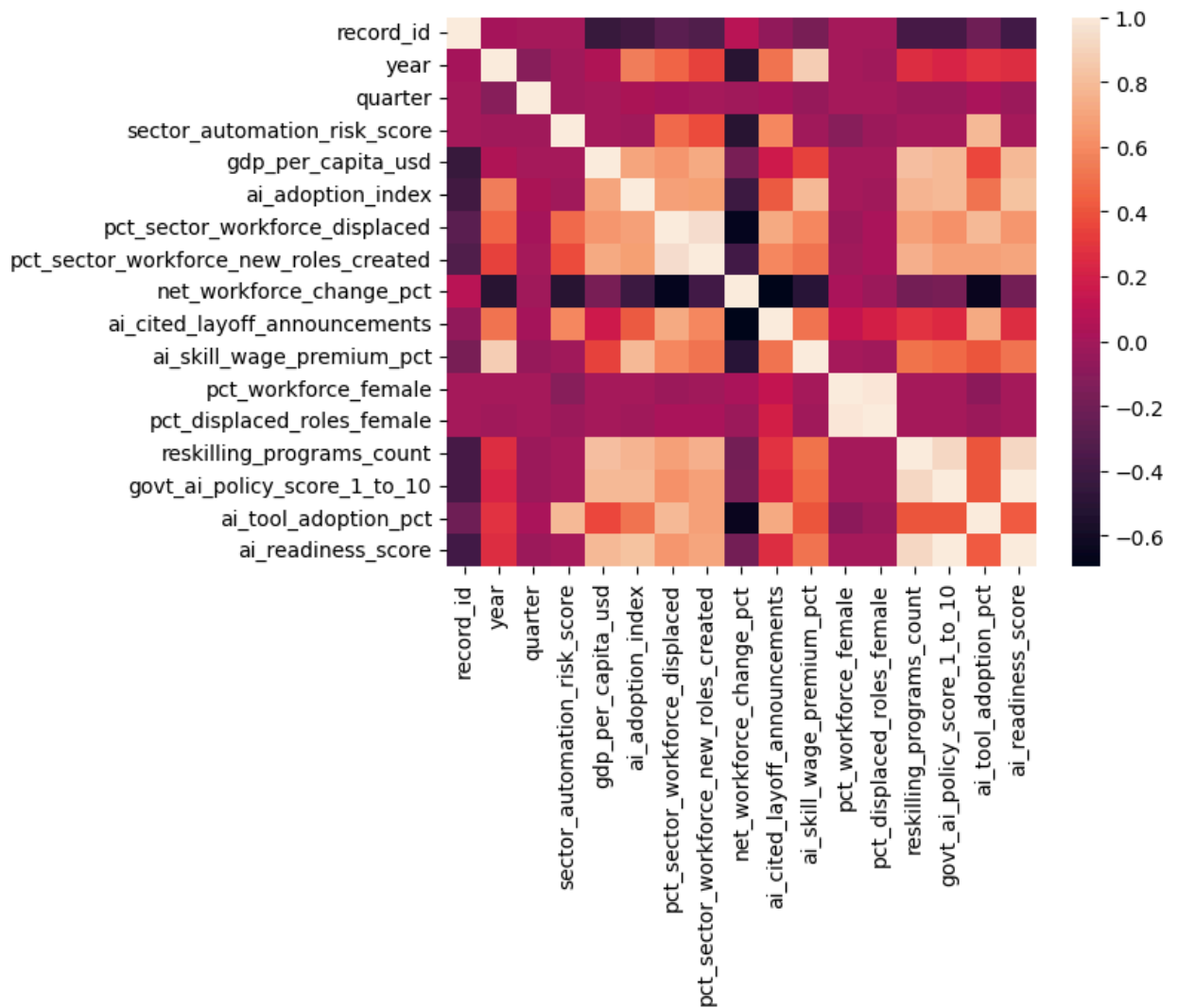


Question 30

Correlation Heatmap

```
In [122...] sns.heatmap(df.corr(numeric_only = True))
```

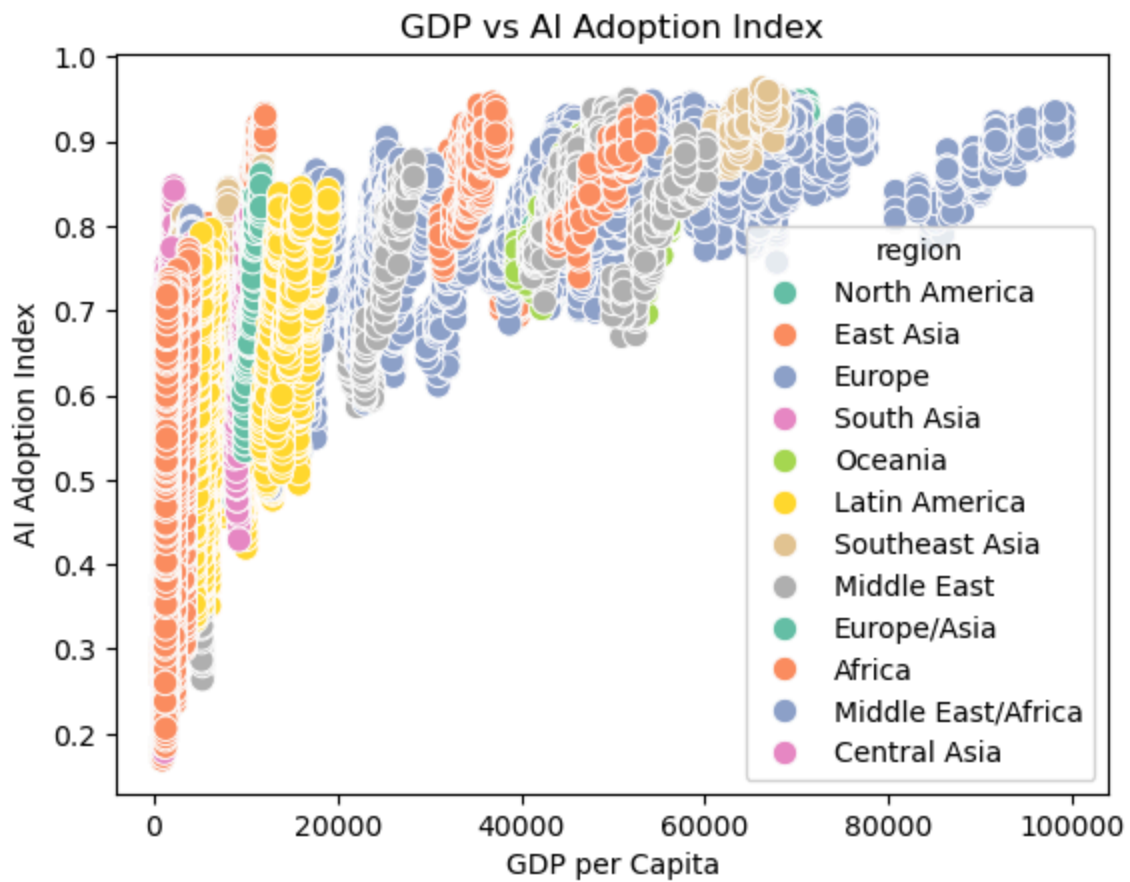
```
Out[122...] <Axes: >
```



Question 31

GDP vs AI Adoption

```
In [125... sns.scatterplot(data = df, x='gdp_per_capita_usd',y='ai_adoption_index', hue='
plt.title("GDP vs AI Adoption Index")
plt.xlabel("GDP per Capita")
plt.ylabel("AI Adoption Index")
plt.show()
```

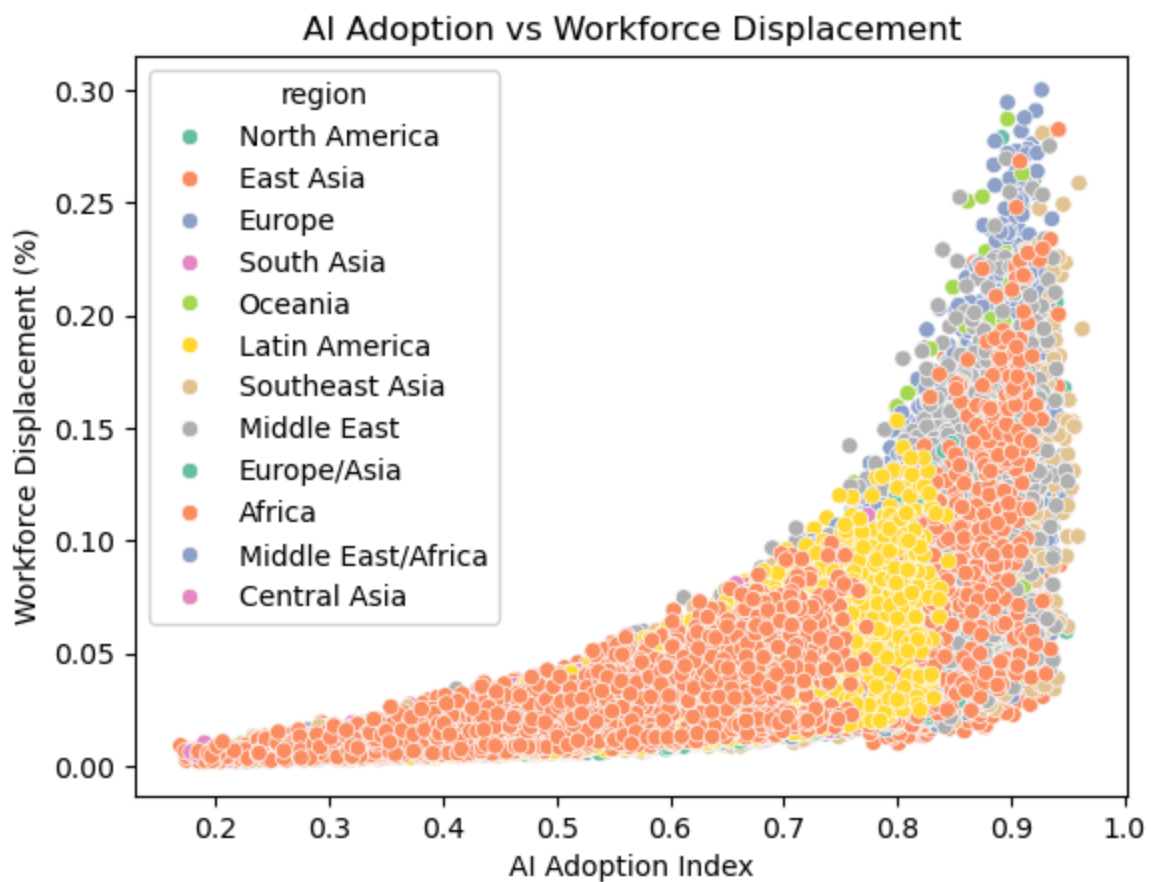



Question 32

AI Adoption vs Workforce Displacement - Scatter Plot

```
In [127... sns.scatterplot(data=df,x='ai_adoption_index',y='pct_sector_workforce_displace

plt.title("AI Adoption vs Workforce Displacement")
plt.xlabel("AI Adoption Index")
plt.ylabel("Workforce Displacement (%)")
plt.show()
```



Question 33

Workforce Displacement by Industry - boxplot

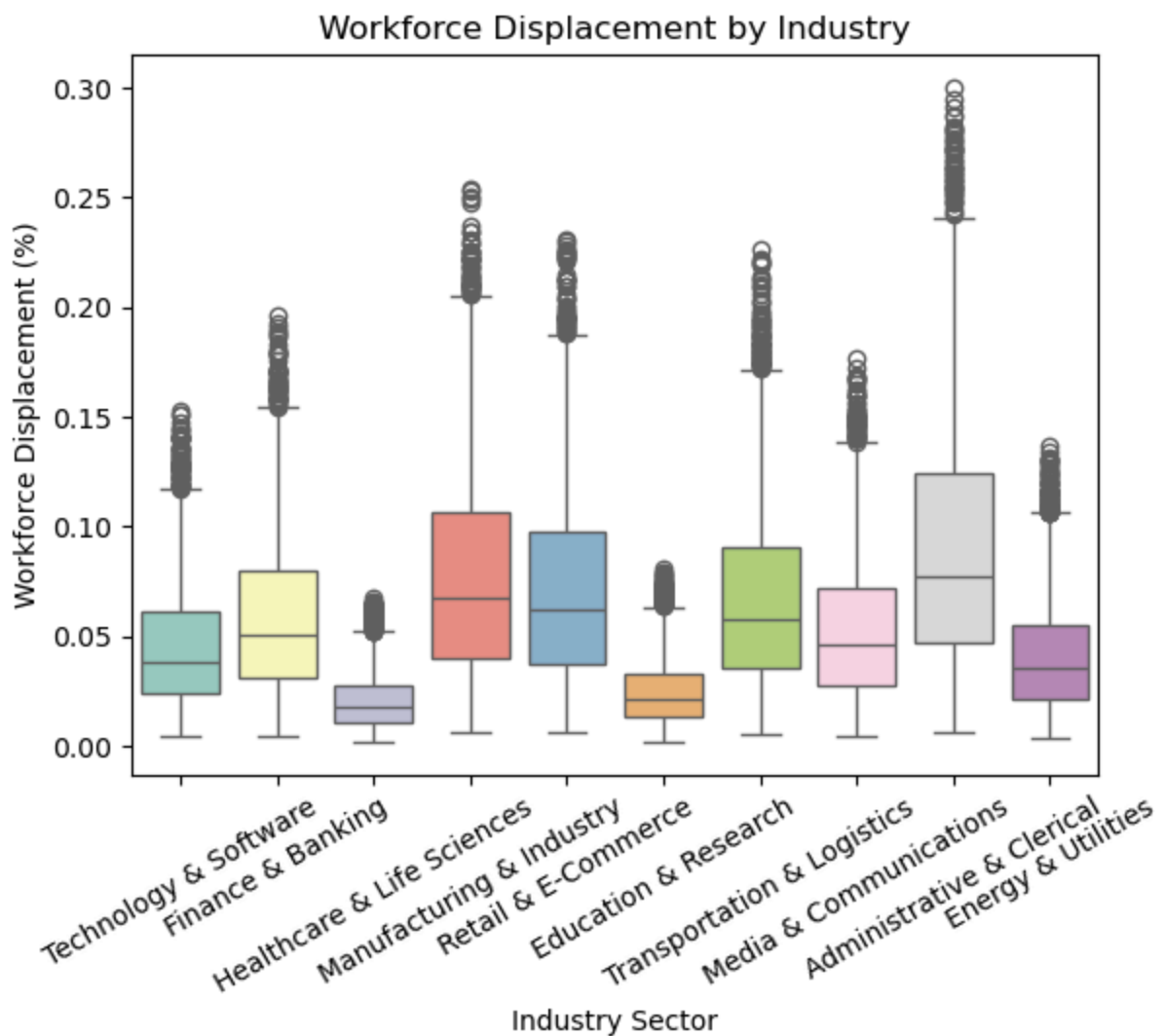
```
In [128... sns.boxplot(
    data=df,
    x='industry_sector',
    y='pct_sector_workforce_displaced',
    palette='Set3'
)

plt.title("Workforce Displacement by Industry")
plt.xlabel("Industry Sector")
plt.ylabel("Workforce Displacement (%)")
plt.xticks(rotation=30)
plt.show()
```

C:\Users\ukti\W\AppData\Local\Temp\ipykernel_8088\24866141.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(
```



Question 34

Automation Risk by Industry (Bar Chart)

```
In [130... # Average automation risk per industry
automation_risk = (
    df.groupby('industry_sector')['sector_automation_risk_score']
      .mean()
      .reset_index()
      .sort_values(by='sector_automation_risk_score', ascending=False)
)

# Bar plot
plt.figure(figsize=(10,6))
sns.barplot(data=automation_risk, x='industry_sector', y='sector_automation_risk_score')

# Add exact value callouts
for i, row in automation_risk.iterrows():
    plt.text(i, row['sector_automation_risk_score']+0.01,
```

```

f"{row['sector_automation_risk_score']:.3f}", ha='center')

plt.title("Automation Risk by Industry")
plt.xlabel("Industry Sector")
plt.ylabel("Average Automation Risk Score")
plt.xticks(rotation=30)
plt.show()

```

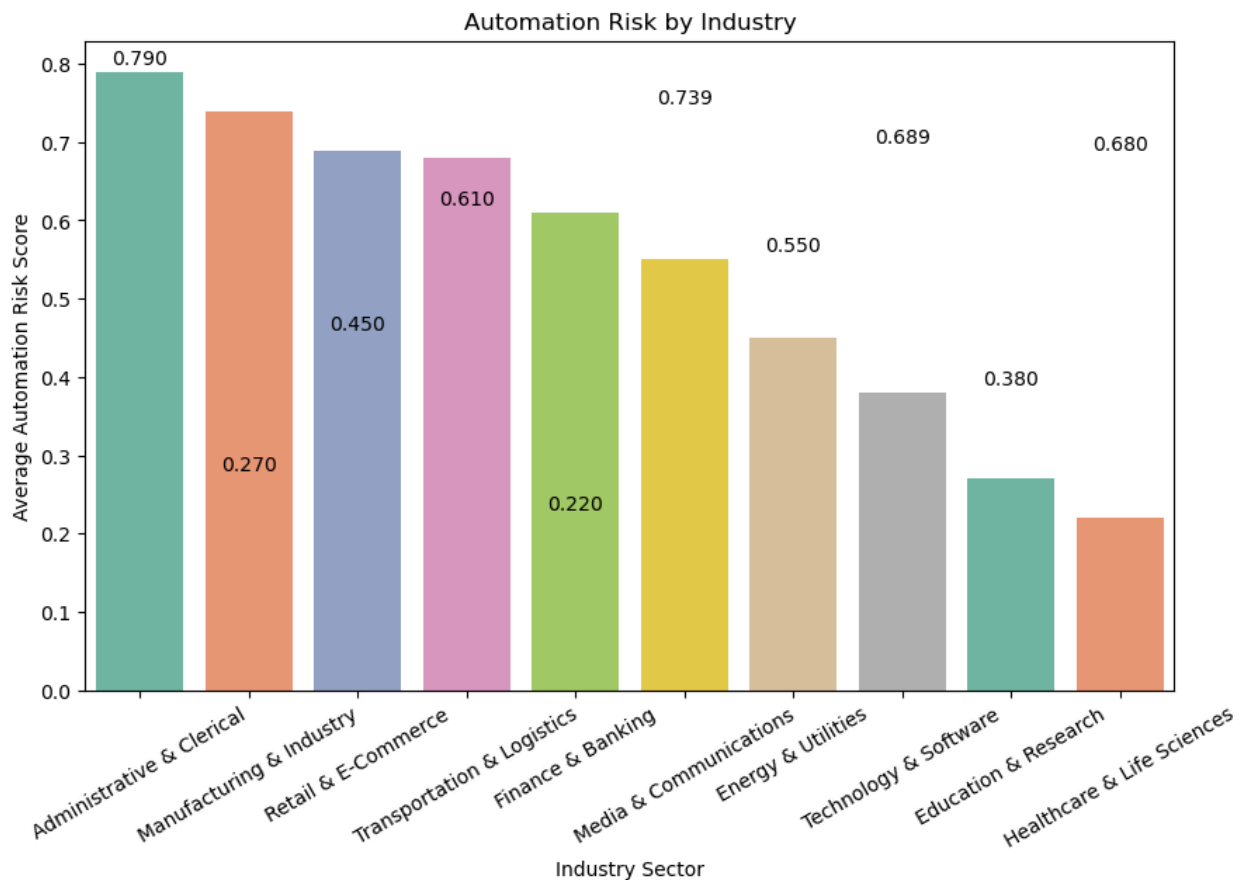
C:\Users\uktiw\AppData\Local\Temp\ipykernel_8088\2410252980.py:11: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```

sns.barplot(data=automation_risk, x='industry_sector', y='sector_automation_risk_score', palette="Set2")

```



Question 35

Top 10 Countries by AI Adoption

```

In [132... top10 = (
    df.groupby('country')['ai_adoption_index']
      .mean()
      .reset_index()

```

```

        .sort_values(by='ai_adoption_index', ascending=False)
        .head(10)
    )

    # Bar plot
    plt.figure(figsize=(10,6))
    sns.barplot(data=top10, x='ai_adoption_index', y='country', palette="Blues_r")

    plt.title("Top 10 Countries by AI Adoption")
    plt.xlabel("Average AI Adoption Index")
    plt.ylabel("Country")
    plt.show()

```

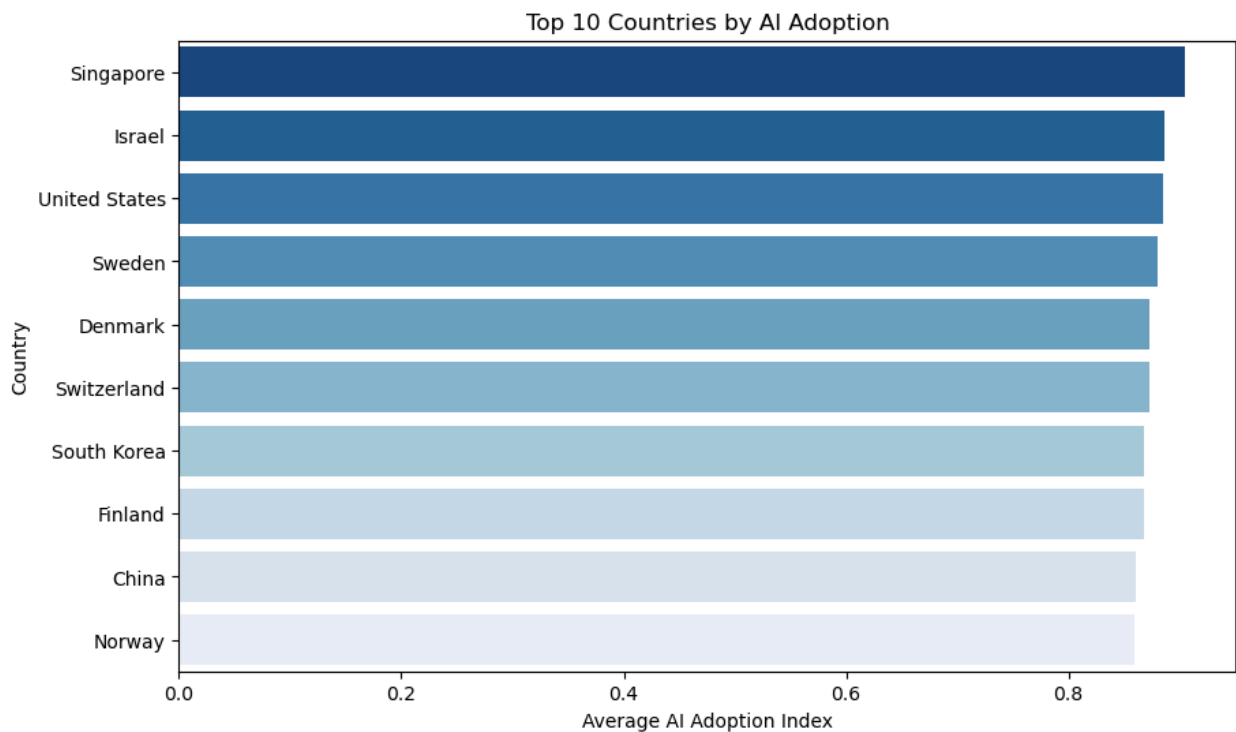
C:\Users\uktiw\AppData\Local\Temp\ipykernel_8088\1091767144.py:11: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```

sns.barplot(data=top10, x='ai_adoption_index', y='country', palette="Blues_r")

```



Question 36

AI Wage Premium by Income Group

```

In [143... df.groupby('income_group')['ai_skill_wage_premium_pct'].mean().plot(
            kind='pie',
            autopct = '%1.1f%%',

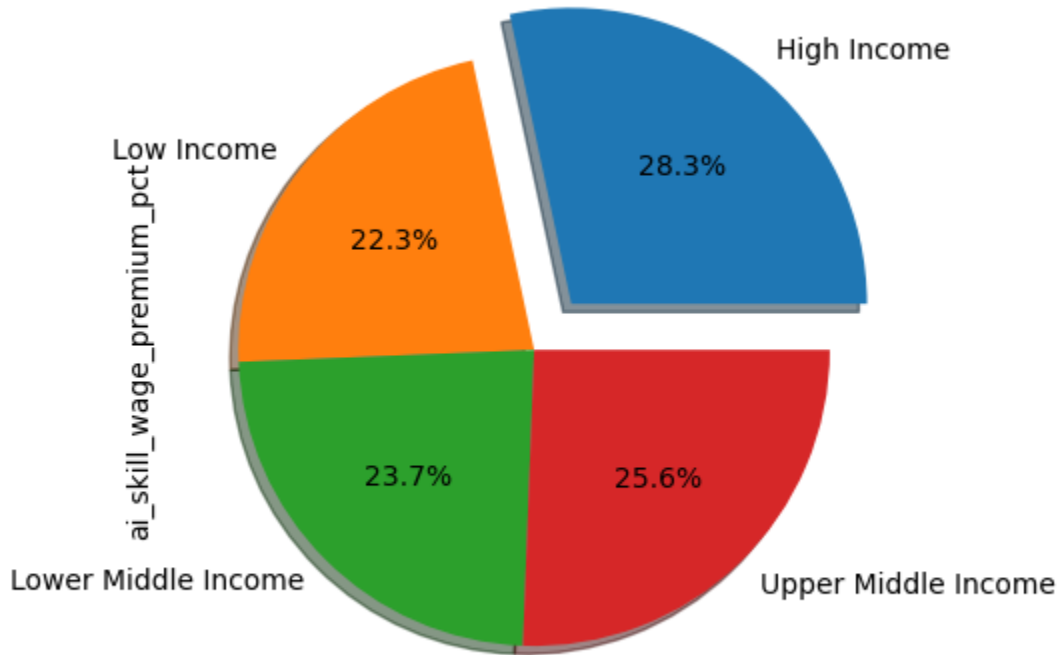
```

```

explode = (0.2,0,0,0),
shadow = True
)

```

Out[143... <Axes: ylabel='ai_skill_wage_premium_pct'>



Question 37

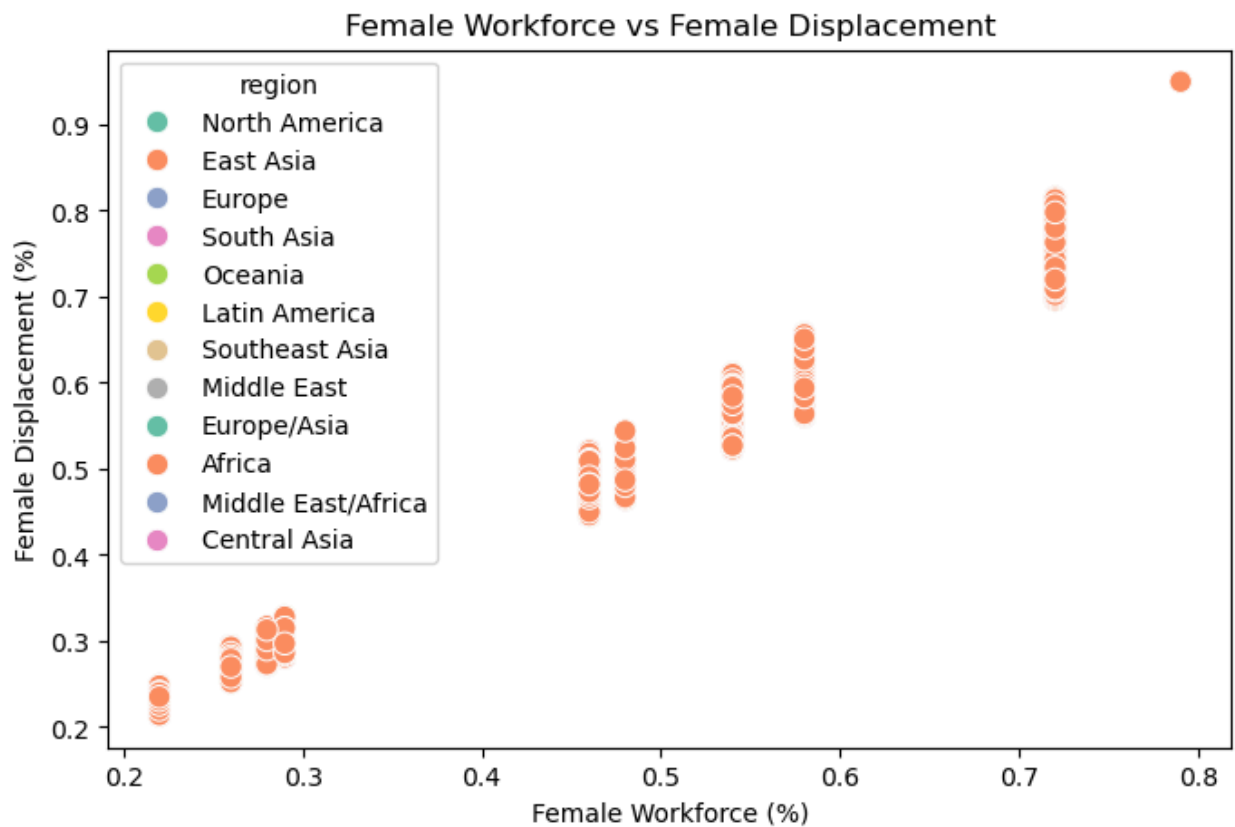
Female Workforce vs Female Displacement

```

In [144... plt.figure(figsize=(8,5))
sns.scatterplot(
    data=df,
    x='pct_workforce_female',
    y='pct_displaced_roles_female',
    hue='region',
    palette='Set2',
    s=80
)

plt.title("Female Workforce vs Female Displacement")
plt.xlabel("Female Workforce (%)")
plt.ylabel("Female Displacement (%)")
plt.show()

```



Question 38

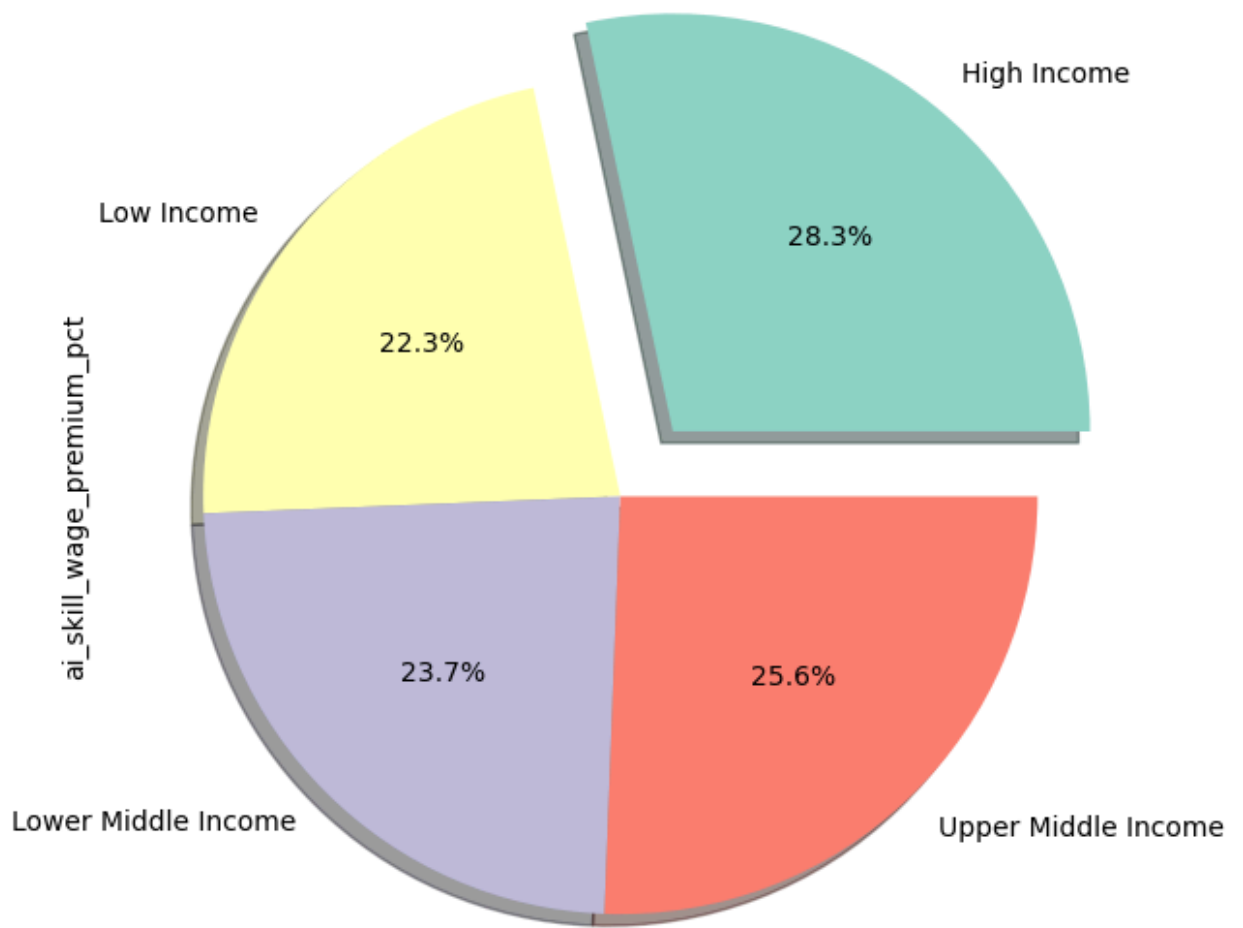
AI Tool Adoption by Income Group

```
In [152... import matplotlib.pyplot as plt

df.groupby('income_group')['ai_skill_wage_premium_pct'].mean().plot(
    kind='pie',
    autopct='%1.1f%%',
    explode=(0.2,0,0,0),
    shadow=True,
    colors=plt.cm.Set3.colors,
    figsize=(7,7),
    title="AI Skill Wage Premium by Income Group"
)
```

```
Out[152... <Axes: title={'center': 'AI Skill Wage Premium by Income Group'}, ylabel='a
i_skill_wage_premium_pct'>
```

AI Skill Wage Premium by Income Group



In []: